Slides created from material by Desmond Elliott and Raquel Fernandez

Multimodal Generative AI Models

Towards Perceptually-grounded NLP Models

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School of informatics

Outline

Aim: Understand why we need Multimodal Models and what are the main design choices that we need to make to design them

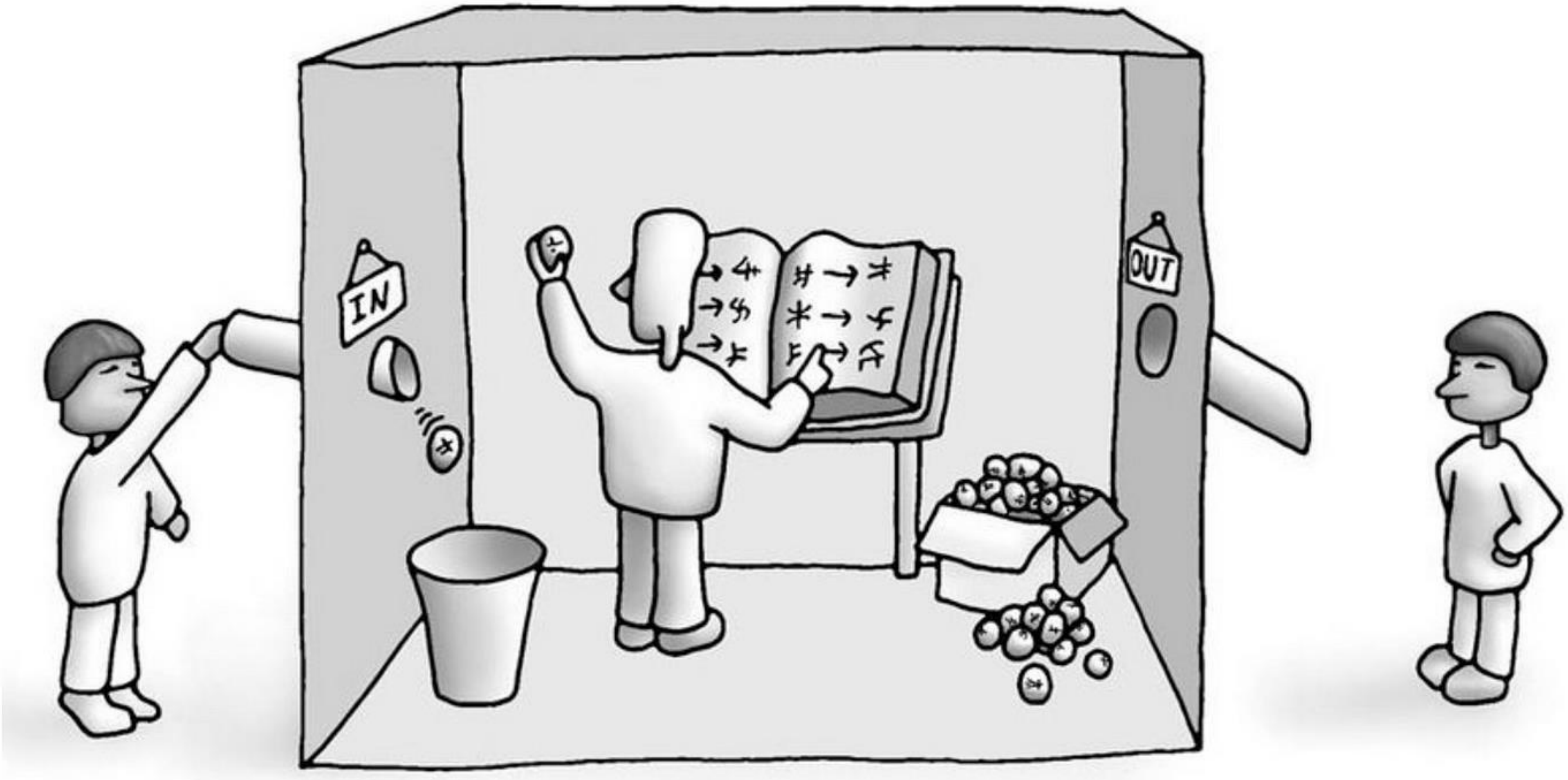
Outline:

- Historical background on Perceptually Grounded NLP
- Vision and Language Models as an implementation of Multimodal Models
  - Input data
  - Model design
  - Training regimes
- Evaluating Vision and Language Models

Grounding – An Overloaded Term

- Grounding in communication (Herbert H. Clark and Susan E. Brennan):
  - *mutual knowledge, mutual beliefs, and mutual assumptions, that is essential for communication between two people*
- In cognitive science and semantics, the symbol grounding problem concerns how it is that words (symbols in general) get their meanings, hence it is concerned with what meaning itself really is (Cangelosi and Harnad)

The Chinese Room Argument



Searle, John R. "Minds, brains, and programs." Behavioral and brain sciences 3.3 (1980): 417-424.

The Chinese/Chinese Dictionary-Go-Round

- Suppose you had to learn Chinese as a second language and the only source of information you had was a Chinese/Chinese dictionary. The trip through the dictionary would amount to a merry-go-round, passing endlessly from one meaningless symbol or symbol-string (the *definientes*) to another (the *definienda*), never coming to a halt on what anything meant
- How do cryptologists of ancient languages and secret code succeed?
  - their efforts are grounded in a first language and in real world experience and knowledge

Searle, John R. "Minds, brains, and programs." Behavioral and brain sciences 3.3 (1980): 417-424.

Symbol Grounding Problem

How can the meanings of the meaningless symbol tokens, manipulated solely on the basis of their (arbitrary) shapes, be grounded in anything but other meaningless symbols?

Harnad, Stevan. "The symbol grounding problem." Physica D: Nonlinear Phenomena 42.1-3 (1990): 335-346.

Symbol Grounding Problem

Amodal Symbols

PASTICCIOTTO

Human Interpretation

Pasticciotto

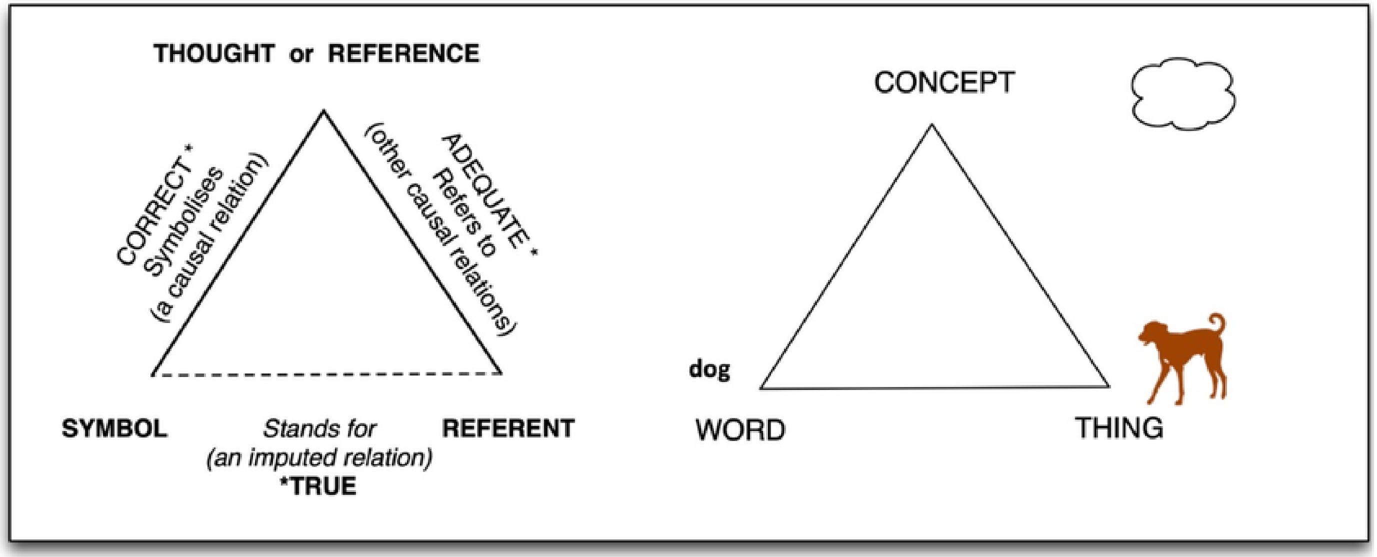
Grounded Symbols

Perception/ Categorisation

Pasticciotto

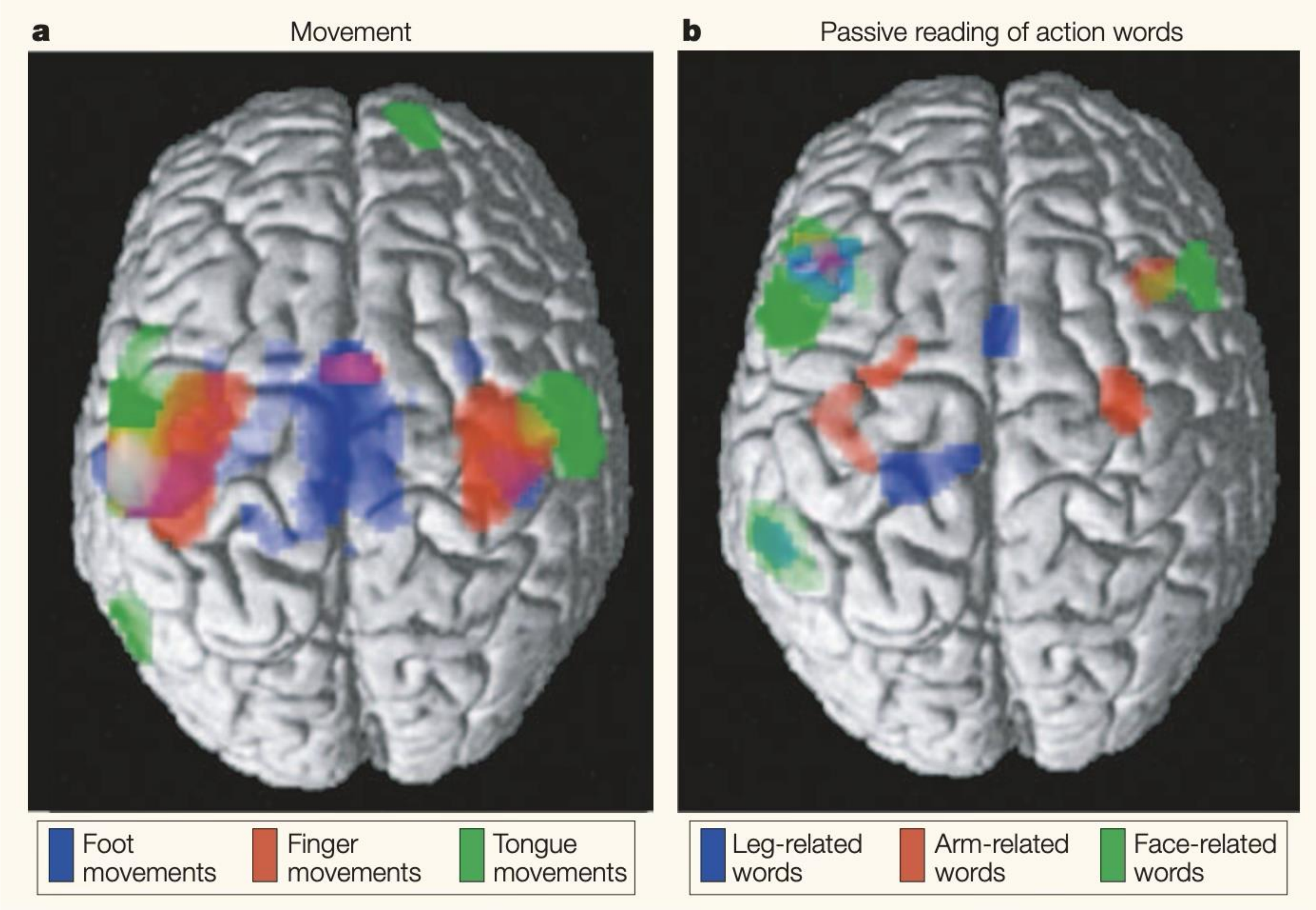
Perceptually Grounded Natural Language Processing

- To be grounded the symbol system would have to be able to pick out the referents of its symbols and its sensorimotor interactions with the world would have to fit coherently with the symbols' interpretations.
- Meaning is multimodal!



Ogden, C. K., & Richards, I. A. (1930). The Meaning of Meaning: A Study of the Influence of Language upon Thought and of the Science of Symbolism. Harcourt, Brace.

Evidence in favour of Multimodal/Grounded NLP



Pulvermüller, F. (2005). Brain mechanisms linking language and action. Nature reviews neuroscience, 6(7), 576-582.

Limits of text-based language representations

- In previous lectures, you've learned that current NLP approaches learn the meaning of words from text only.

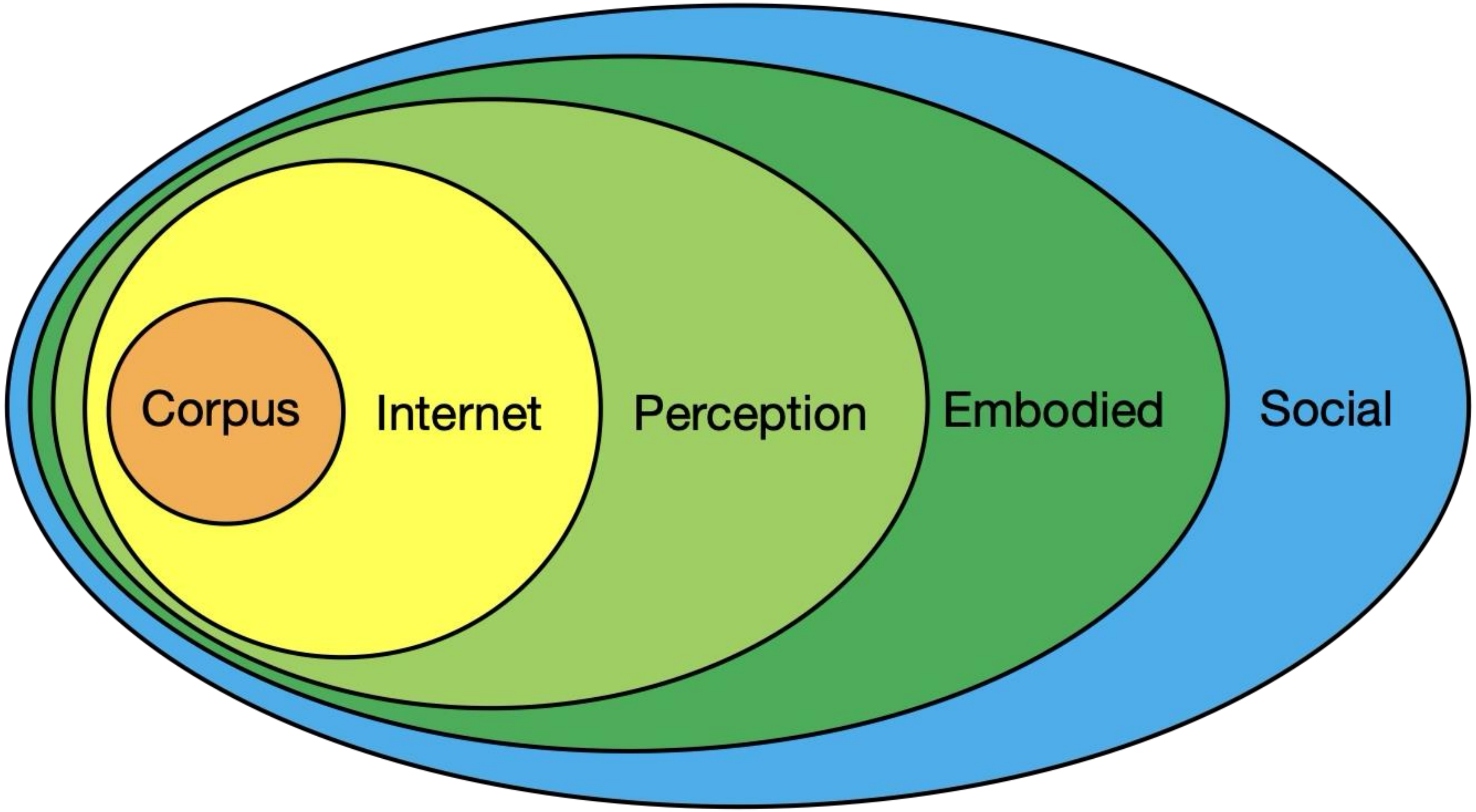
- But this contradicts the theoretical evidence discussed in this lecture and raises a few important questions:
  1. What are the limits of text-based representations?
  2. Should we be learning this way?
  3. Is it data-efficient/effective?
  4. Why should models learn this way, but not humans?

LLM’s answer to this problem

- We can just increase data and model size to get better models
- When trained with specific types of data (e.g., instructions), they can perform complex NLP tasks such as question answering

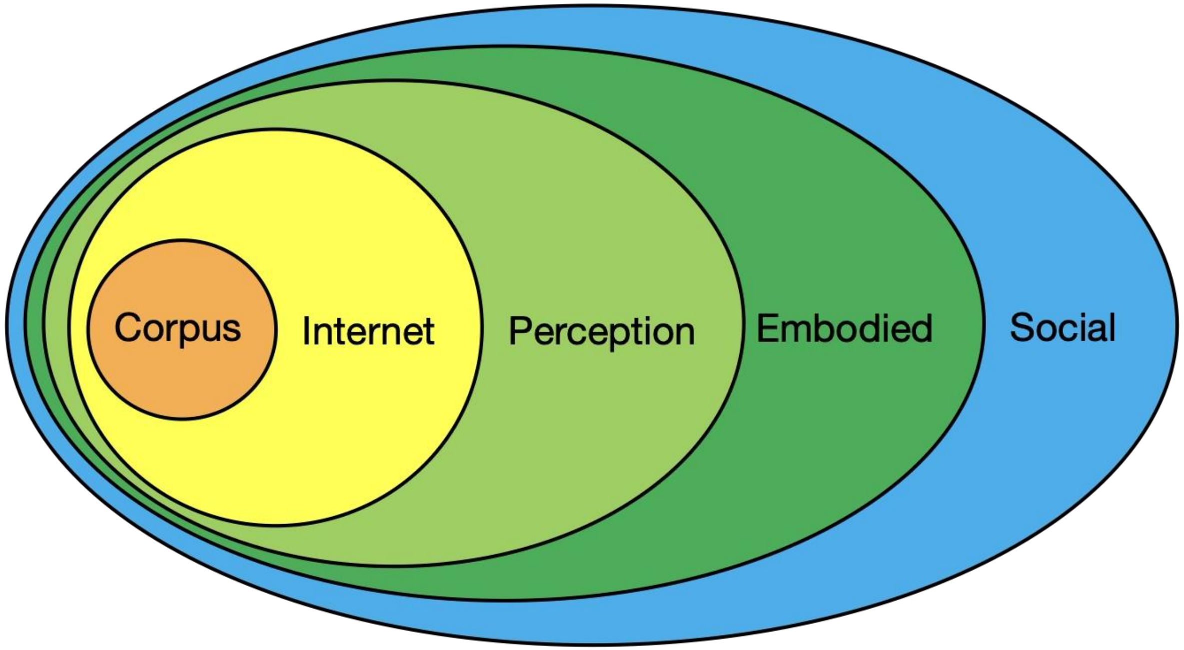
- However, they are trained with trillions of tokens making them extremely inefficient learners
    - What can we do to improve on this?

Experience Grounds Language



Bisk, Y., Holtzman, A., Thomason, J., Andreas, J., Bengio, Y., Chai, J., ... & Turian, J. (2020, November). Experience Grounds Language. EMNLP 2020

Experience Grounds Language

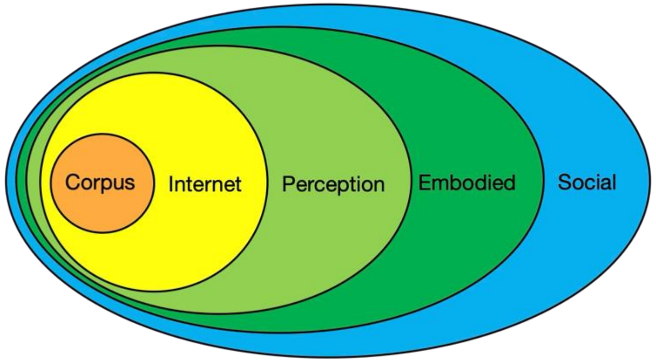


Each layer is a world scope = Describes the "data" that the learner has access to

How does the reporting bias change?
What new knowledge do models have access to?
What advances in modelling and fusion are necessary?

Bisk, Y., Holtzman, A., Thomason, J., Andreas, J., Bengio, Y., Chai, J., ... & Turian, J. (2020, November). Experience Grounds Language. EMNLP 2020

What delimits a World Scope?



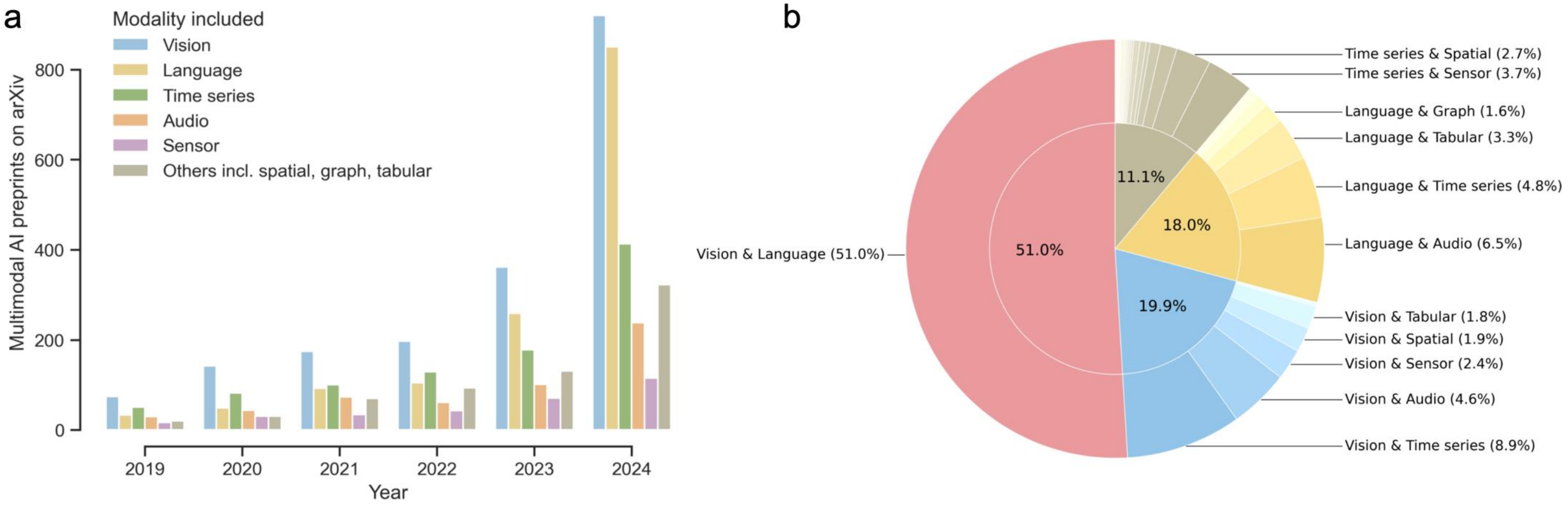
You can't learn language:

- **from the radio (Internet)**. $WS_2 \subset WS_3$
  - A learner cannot be said to be in WS3 if it can succeed without perception (e.g., visual, auditory).
- **from the television.** $WS_3 \subset WS_4$
  - A learner cannot be said to be in WS4 if the space of its world actions and consequences can be enumerated.
- **by yourself.** $WS_4 \subset WS_5$
  - A learner cannot be said to be in WS5 unless achieving its goals requires cooperating with a human in the loop.

Bisk, Y., Holtzman, A., Thomason, J., Andreas, J., Bengio, Y., Chai, J., ... & Turian, J. (2020, November). Experience Grounds Language. EMNLP 2020

Multimodal Artificial Intelligence

Multimodal Artificial Intelligence (AI) integrates diverse types of data via machine learning to improve understanding, prediction, and decision-making across disciplines such as healthcare, science, and engineering.

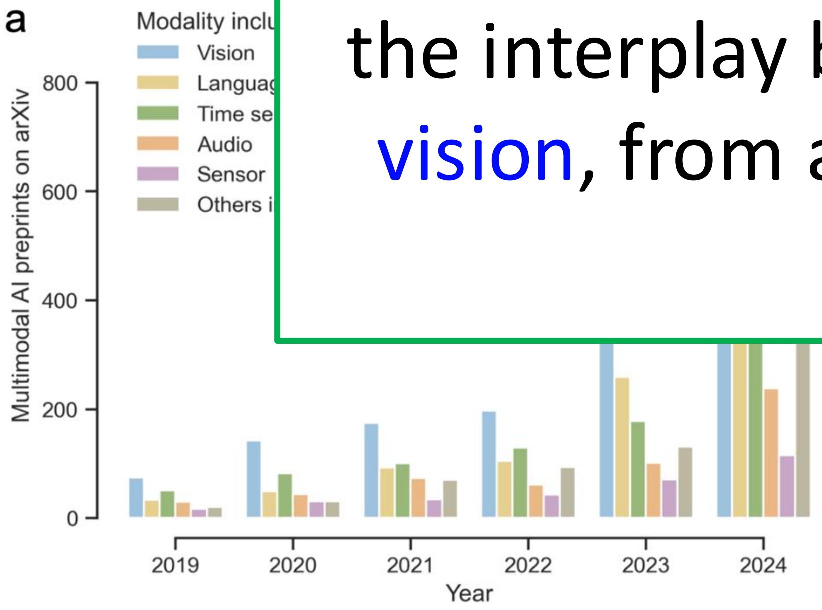


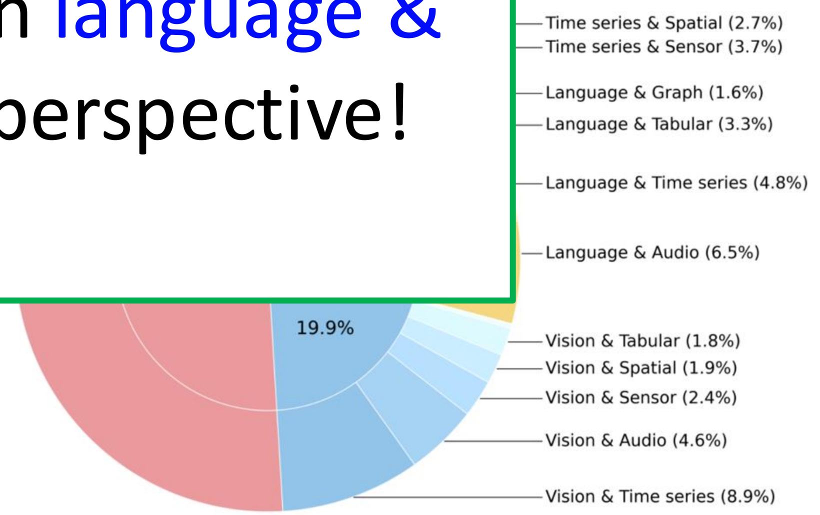
Liu, Xianyuan, et al. "Towards deployment-centric multimodal AI beyond vision and language." arXiv preprint arXiv:2504.03603 (2025).

Multimodal Artificial Intelligence

Multimodal Artificial Intelligence (AI) integrates diverse types of data via machine learning to improve understanding, prediction, and decisio…re, science,

In these lectures, we will focus on the interplay between language & vision, from an NLP perspective!





Liu, Xianyuan, et al. "Towards deployment-centric multimodal AI beyond vision and language." arXiv preprint arXiv:2504.03603 (2025).

Vision+Language Models

A Vision and Language Model integrates vision and language data via machine learning to improve understanding, prediction, and decision-making across many disciplines.

A group of people eating noodles

Vision+Language Models Tasks

- Image captioning
  - A group of people eating noodles
- Visual Question Answering
  - Q: What are the people eating?
  - A: Noodles
- Visual grounding
  - Given a language description, identifies the associated target object

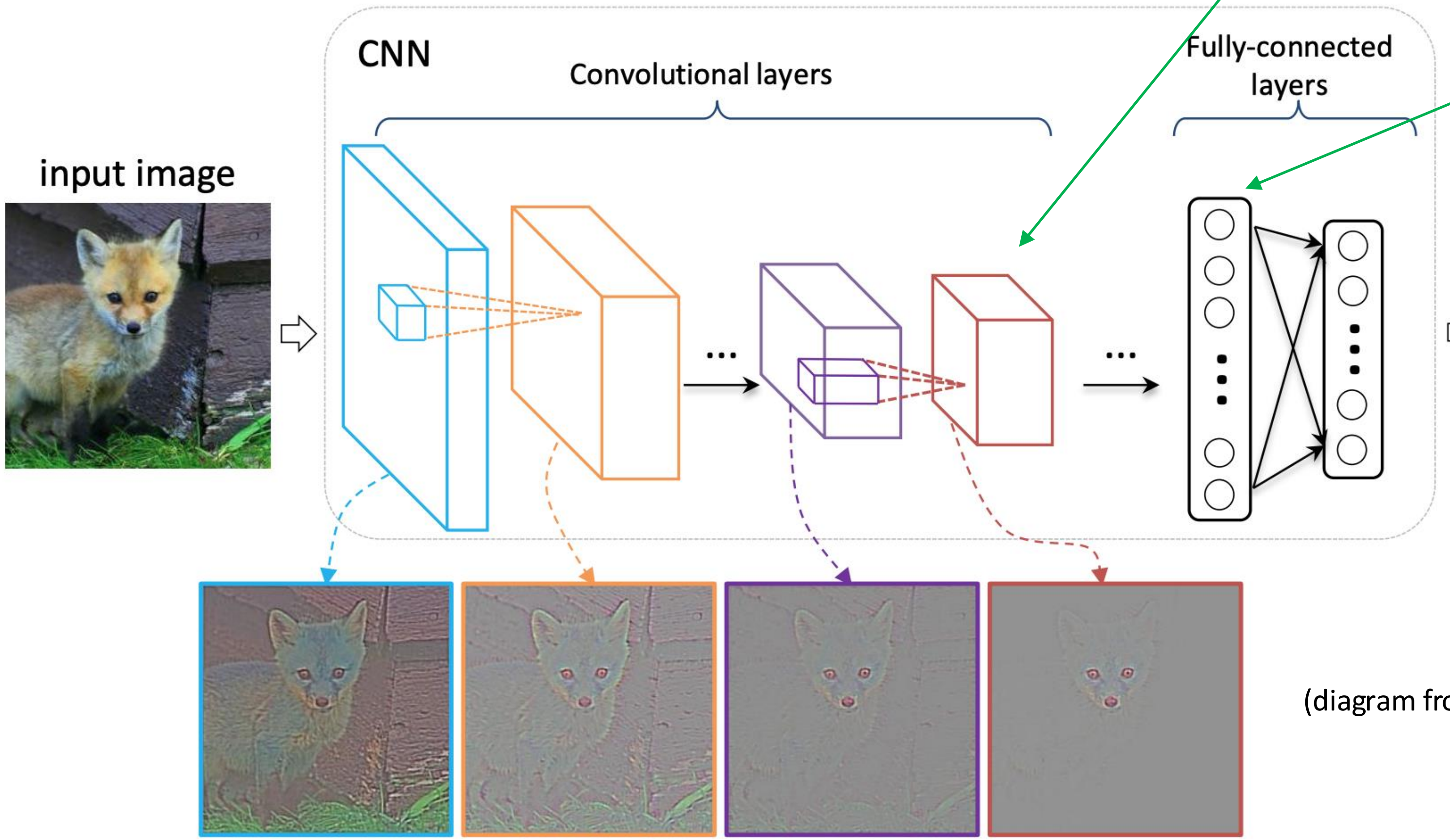
Representing visual information

You have already seen how to learn text representations (see ANLP and previous lectures)

How do we represent information in other modalities, in particular vision?

- In very early approaches, symbolic features were used to represent objects or scenes, without any vision
- As computer vision methods started to be further developed, the focus shifted towards automatically learning to represent visual information -> see Computer Vision [INFR112122024]

Convolutional Neural Networks (CNNs)

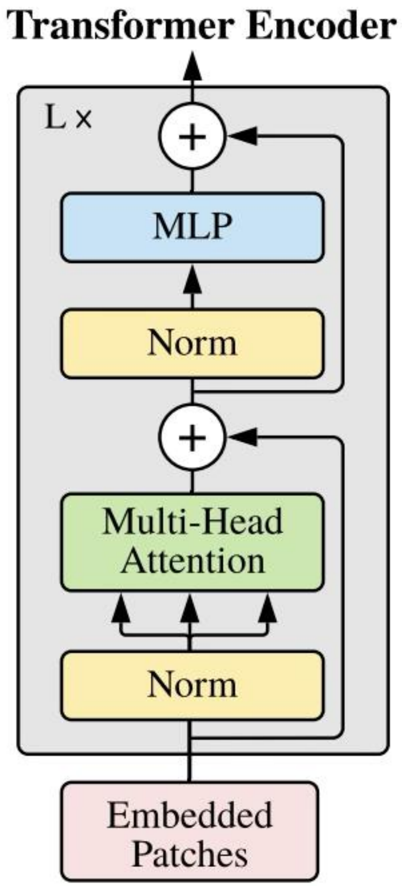


class name	prob
kit fox	0.5956
red fox	0.3576
grey fox	0.0439
coyote	0.0013
Arctic fox	0.0003

(diagram from Yu et al. ICLR 2015)

Simonyan, Karen, and Andrew Zisserman. "Very deep convolutional networks for large-scale image recognition." arXiv preprint arXiv:1409.1556 (2014).

Vision Transformers



Dosovitskiy, Alexey. "An image is worth 16x16 words: Transformers for image recognition at scale." arXiv preprint arXiv:2010.11929 (2020).
Image credit: https://www.v7labs.com/blog/vision-transformer-guide

Task-specific Vision+Language Models

DESIGNING VISION+LANGUAGE MODELS

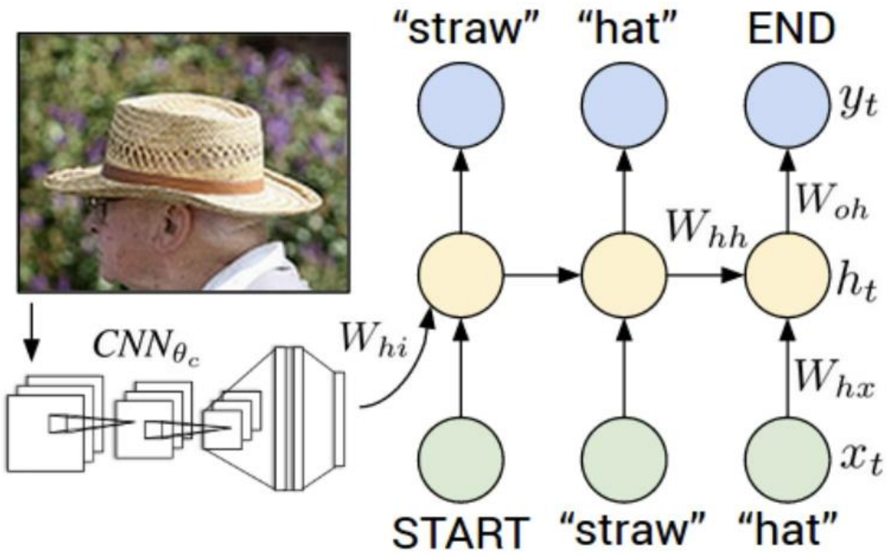
Task-specific models: Image captioning

A language model conditioned on visual information implemented as an encoder-decoder neural architecture

Model: uses a CNN to extract a pooled representation that it is used to initialise the hidden state of a language model RNN

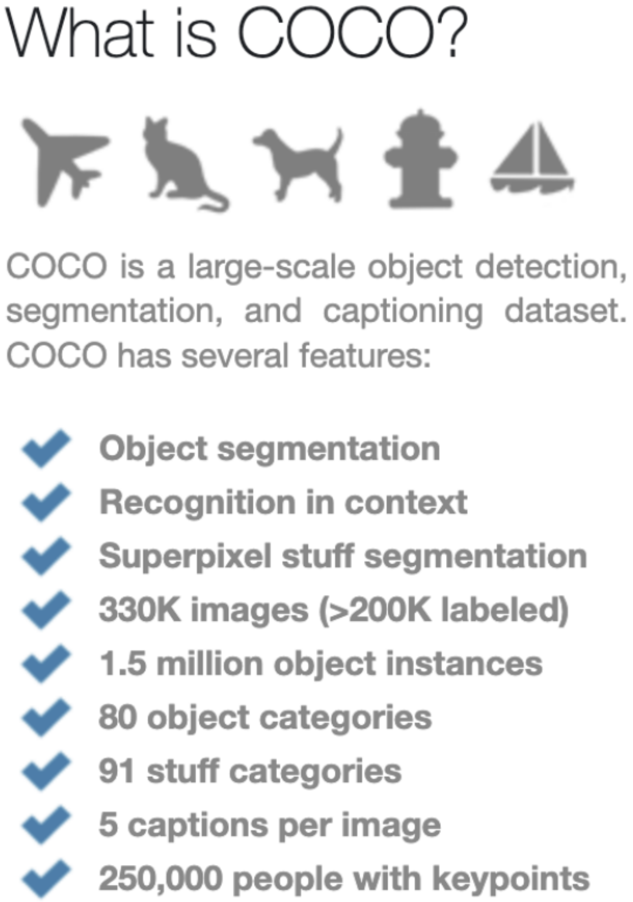
- The RNN generates tokens one by one conditioned on the previous tokens
- Note: conditioning on the image happens only at the very beginning of the sequence

- This could be problematic; why?



Karpathy, A., & Fei-Fei, L. (2015). Deep visual-semantic alignments for generating image descriptions. In Proceedings of the IEEE conference on computer vision and pattern recognition (pp. 3128-3137).

Image captioning datasets -- MSCOCO

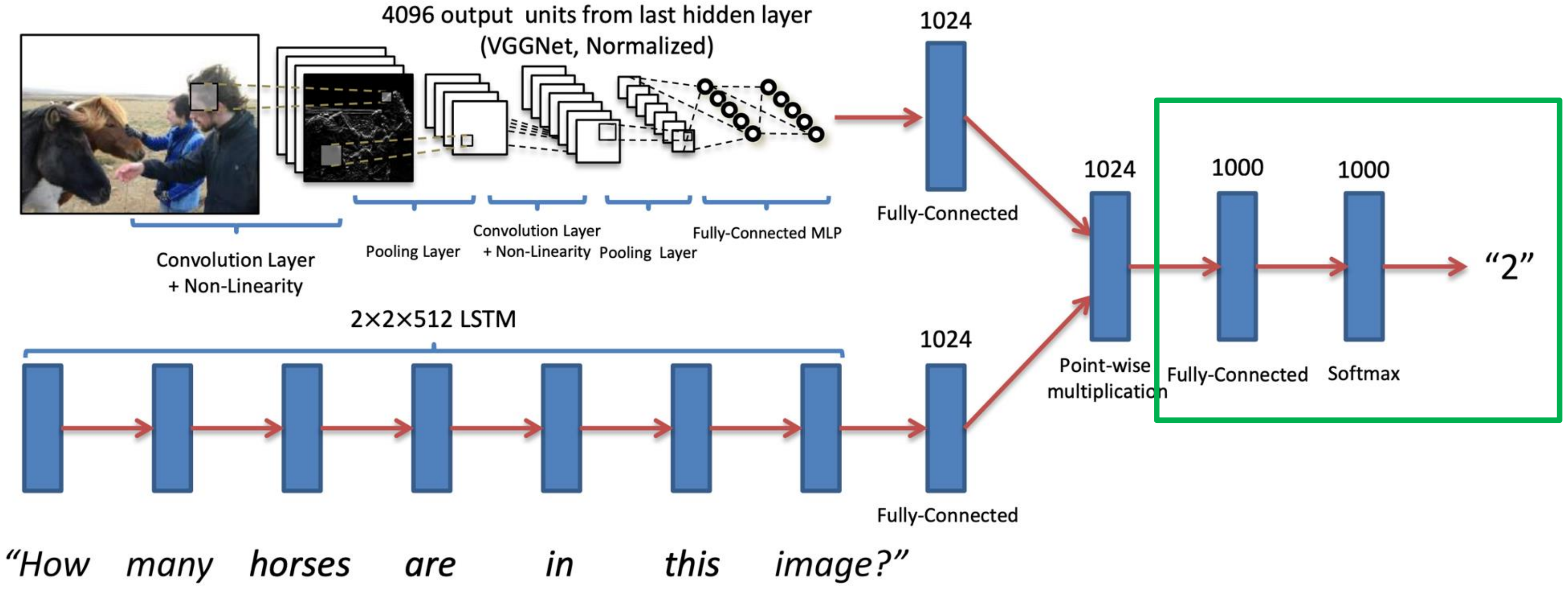


dounts are on sale under a glass display.
a row of green donuts, a row of glazed donuts, and a row of chocolate donuts
a variety of doughnuts on display in a bakery window.
there are three rows of different kinds of donuts
the different flavored donuts are on display in the glass case.

Karpathy, A., & Fei-Fei, L. (2015). Deep visual-semantic alignments for generating image descriptions. In Proceedings of the IEEE conference on computer vision and pattern recognition (pp. 3128-3137).

Task-specific models: Visual Question Answering (VQA)

Similar architecture as before but with a dedicated classification head for answer prediction:



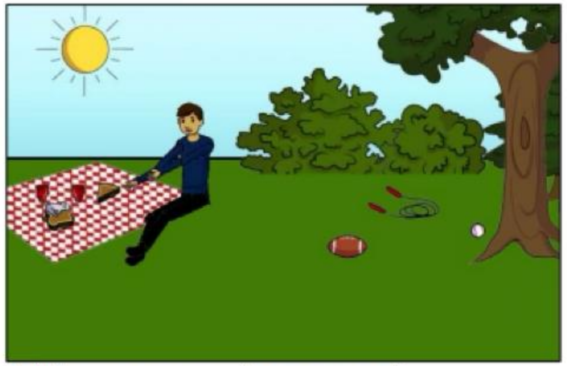
Antol et al. (2015). VQA: Visual Question Answering. ICCV.

Visual question answering datasets – VQA dataset

- Answer questions about real-world images scraped from the web
- Multimodal inputs: Image & Question
- Originally tackled as a classification task: each answer is a class
- VQA dataset: around 600k image-question pairs

What color are her eyes?
What is the mustache made of?

How many slices of pizza are there?
Is this a vegetarian pizza?



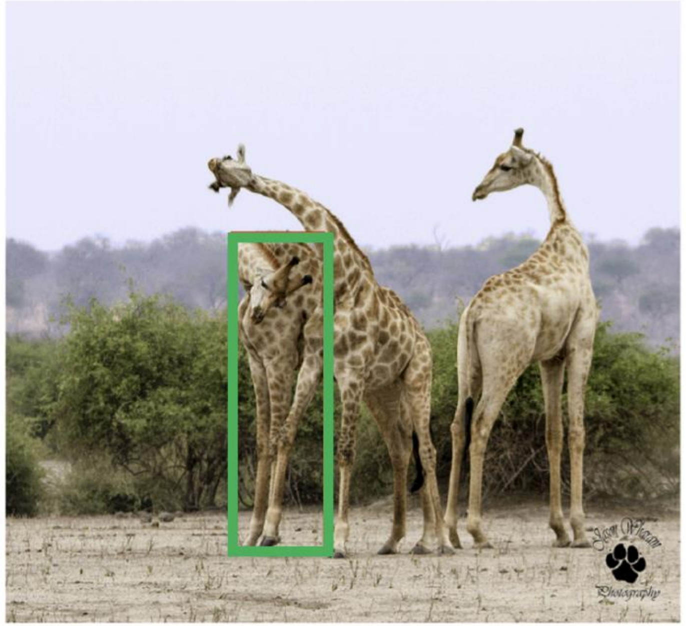
Is this person expecting company?
What is just under the tree?

Does it appear to be rainy?
Does this person have 20/20 vision?

Antol et al. (2015). VQA: Visual Question Answering. ICCV.

Referential Expression Generation and Understanding

- An essential task for assessing models' visual grounding abilities
- A relevant task for many human-robot collaboration tasks
  - E.g.: "Pass me the blue mug on the shelf"
- Implemented as two complementary tasks:
  - Generation: generate description corresponding to a bounding box
  - Understanding: select/predict a bounding box that matches a given textual description



RefCOCO:
1. giraffe on left
2. first giraffe on left

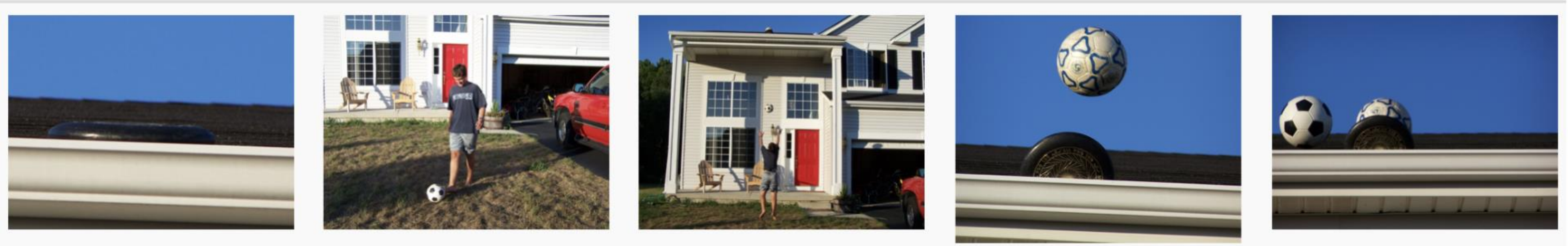
RefCOCO+:
1. giraffe with lowered head
2. giraffe head down

RefCOCOg:
1. an adult giraffe scratching its back with its horn
2. giraffe hugging another giraffe

Yu, Licheng, et al. "Modeling context in referring expressions." ECCV 2016.

Visual Storytelling (VIST) datasets

- "Storyable" events tend to involve some form of possession, e.g., "John's birthday party"; use this as an extraction rule!
- 5 images from the same Flickr album (around 20k sequences in total)
- Crowdsourced stories: one sentence per image; several stories per image sequence



	1	2	3	4	5
Image descriptions in isolation	A black frisbee is sitting on top of a roof.	A man playing soccer outside of a white house with a red door.	The boy is throwing a soccer ball by the red door.	A soccer ball is over a roof by a frisbee in a rain gutter.	Two balls and a frisbee are on top of a roof.
Sequence description (story)	**A discus got stuck up on the roof.**	**Why not try getting it down with a soccer ball?**	**Up the soccer ball goes.**	**It didn't work so we tried a volley ball.**	**Now the discus, soccer ball, and volleyball are all stuck on the roof.**

Huang, Ting-Hao, et al. "Visual storytelling." Proceedings of the 2016 conference of the North American chapter of the association for computational linguistics: Human language technologies. 2016.

Visual dialogue datasets – GuessWhat?!

Visually grounded game with two players:

1. Guesser: selects an object in the image as target object and answers questions with yes/no only
2. Questioner: generates questions that are useful to guess the target object

Crowdsourced 150K dialogues using 66k MSCOCO images

Questioner	Oracle
Is it a vase?	Yes
Is it partially visible?	No
Is it in the left corner?	No
Is it the turquoise and purple one?	Yes

De Vries, Harm, et al. "Guesswhat?! visual object discovery through multi-modal dialogue." Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition. 2017.

General-purpose Vision+Language models

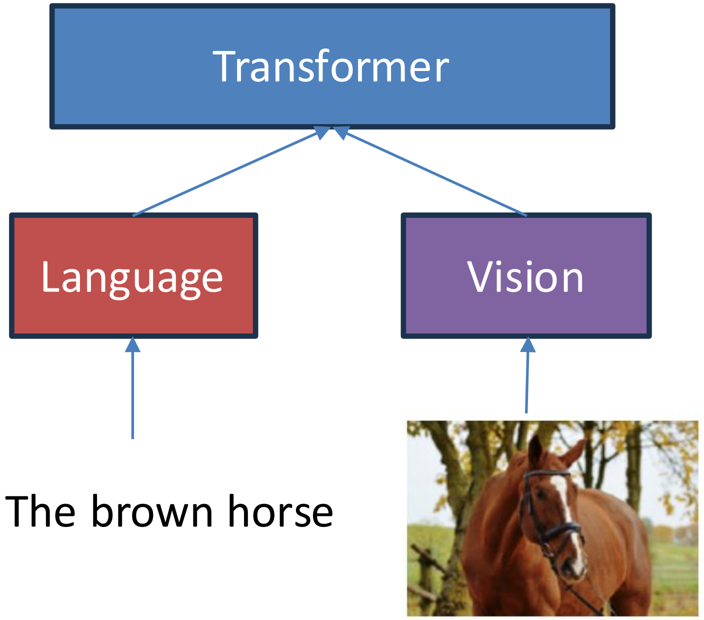
DESIGNING VISION+LANGUAGE MODELS

Main Approaches for Designing Transformer-based VLMs

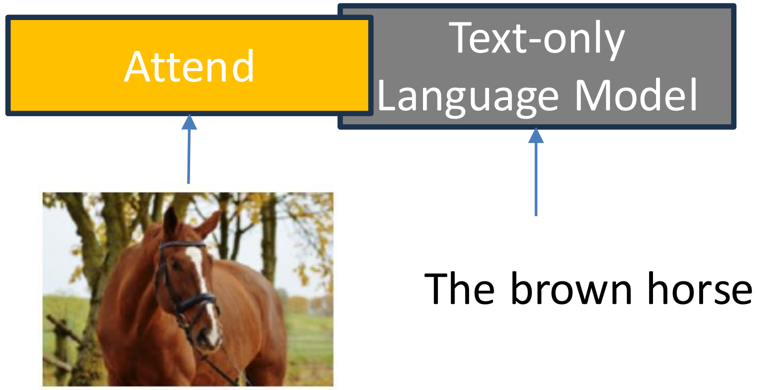
Most popular VLM architectures rely on Transformers which operate on discrete tokens. In the literature we have different ways to facilitate to fuse the language and visual modalities:

Next lecture!

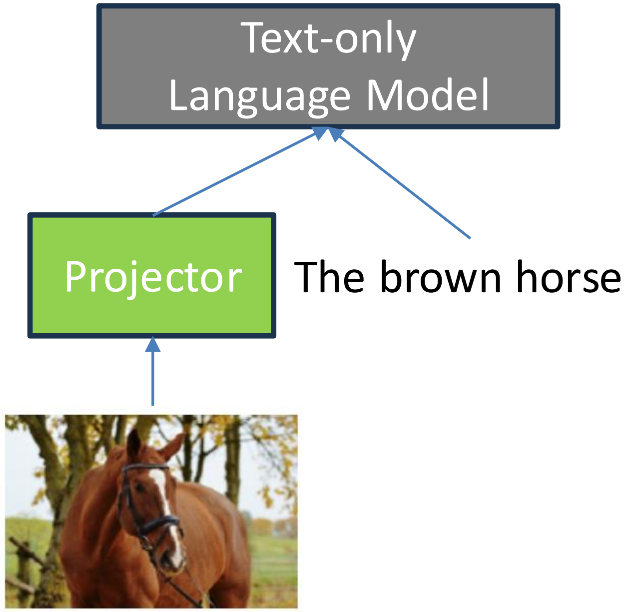
Cross-encoding



Cross-attention

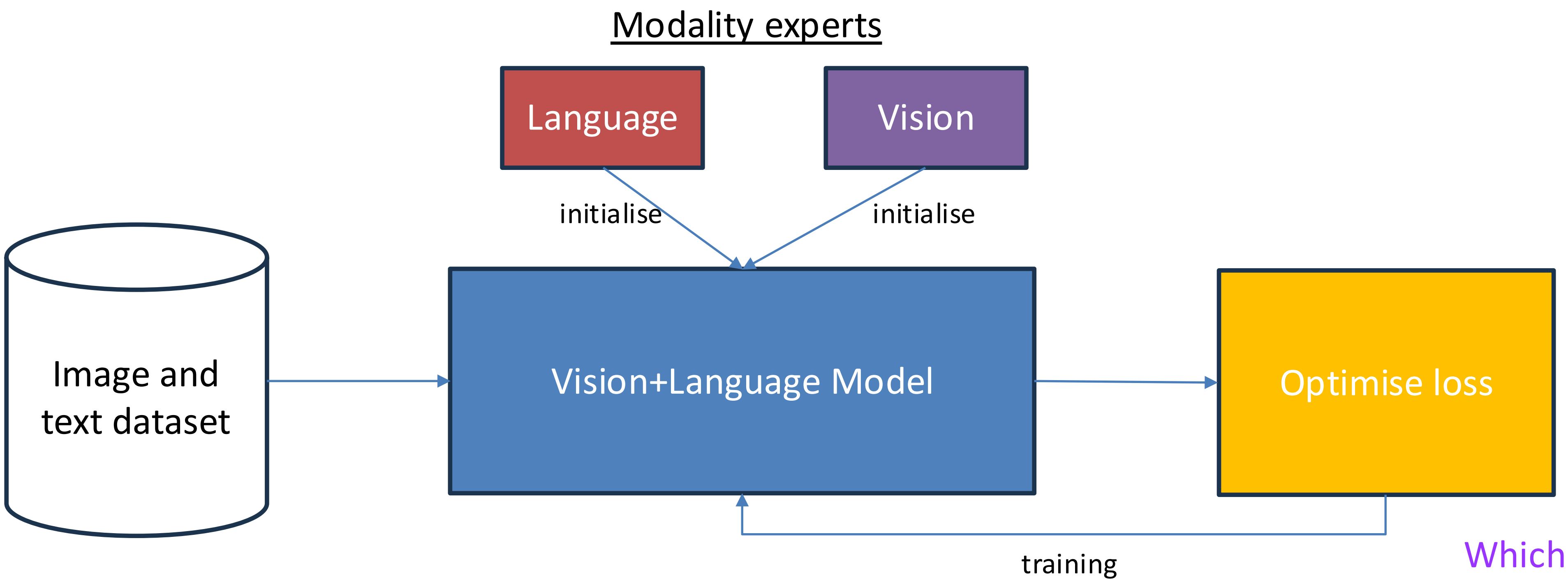


Visual prefix



Cross-encoding: Bugliarello et al. 2021. Multimodal Pretraining Unmasked: A Meta-Analysis and a Unified Framework of Vision-and-Language BERTs. TACL.
Cross-attention: Alayrac, Jean-Baptiste, et al. "Flamingo: a visual language model for few-shot learning." NeurIPS (2022).

Training pipeline



Which loss?

Data for Pretraining Vision+Language Models

What could be a reasonably sized dataset for V+L pretraining?

- Internet-scale data are essential for model pretraining
  - ✓ Idea 1: alt-caption available on websites as *noisy* captions
  - ✓ Idea 2: use web pages with interleaved text and images
  - ✓ Idea 3: use photos available on <your favourite social media>
- This type of data are important because they will teach the model the grounding between the vision and language modality
- Does it matter that they are potentially noisy?
  - Not really; real world data are messy!

Singh, A., Hu, R., Goswami, V., Couairon, G., Galuba, W., Rohrbach, M., & Kiela, D. (2022). Flava: A foundational language and vision alignment model. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (pp. 15638-15650).

Training losses for V+L models

Optimise loss

- L_{MIM}: Masked Image Modelling = Randomly mask some patches in the image, and ask to reconstruct them (e.g., predict a tensor of the same size of the vision tokens and use MSE loss)
- L_{LM}: Language Modelling = Classic language modelling loss or masked language modelling (like BERT)
- $L_{\mathrm{IT}M}$: Image-Text Modelling = Given an image and a caption, decide whether the image matches the caption (i.e., binary classification)
- $L_{M\mathrm{M}M}$: Masked Multimodal Modeling = Simultaneously mask both vision and language, and predict both of them

Singh, A., Hu, R., Goswami, V., Couairon, G., Galuba, W., Rohrbach, M., & Kiela, D. (2022). Flava: A foundational language and vision alignment model. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (pp. 15638-15650).

From task-specific performance to representational quality

EVALUATING VISION+LANGUAGE MODELS

Extrinsic Evaluation

Definition: a task/dataset aimed at assessing the performance of a VLM in a specific task that is relevant for real world applications

Historically, practitioners have focused on two types of evaluation setups:

Task-specific evaluation	Multi-task evaluation
+ Focused on a specific capability + Very precise evaluation metrics + Direct application to use case ✗ Results not transferable	+ Wide range of language abilities + Wide range of visual abilities + Results transferable to other tasks ✗ Hard to aggregate results from different tasks

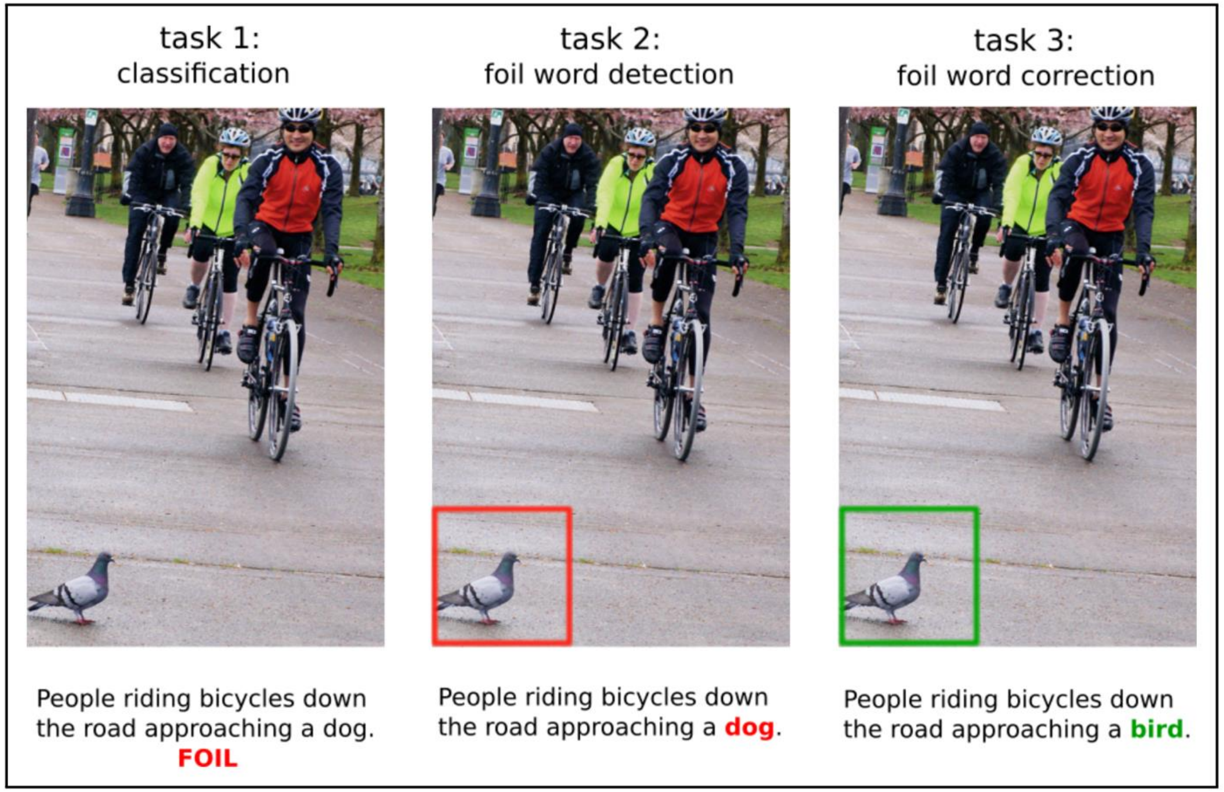
Task-specific Evaluation

- By grounding language into vision, arguably multimodal models have a representational advantage over text-only models
- We therefore need ways to evaluate the quality of the representations learned by the models
- This was traditionally done by assessing very specific capabilities of the models that are relevant for perceptually grounded tasks

Challenge datasets to analyse specific skills

Early example: FOIL captions

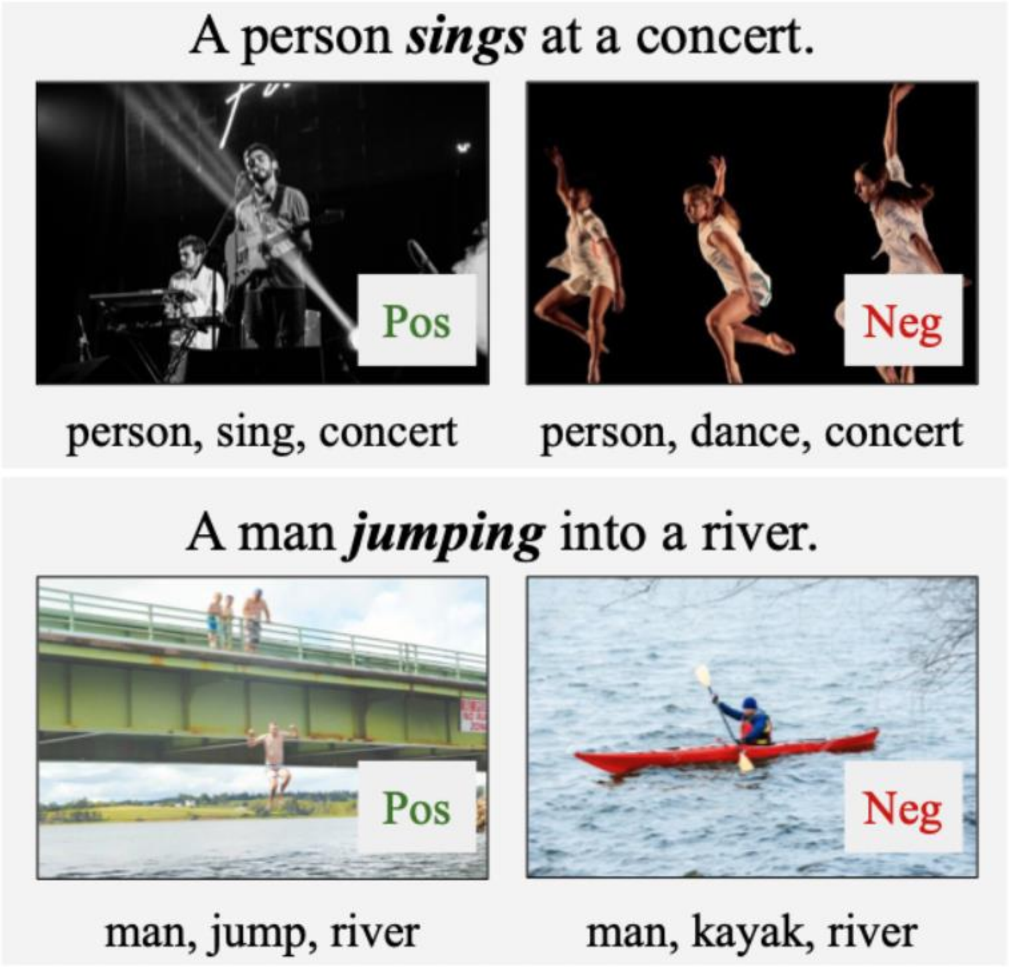
- Do V&L models really understand the relationship between words and images?
- Crowdsource datasets that contain contextually plausible but incorrect image-text pairs, focusing on nouns



Ravi Shekhar, Sandro Pezzelle, Yauhen Klimovich, Aurélie Herbelot, Moin Nabi, Enver Sangineto, and Raffaella Bernardi. "FOIL it! Find One mismatch between Image and Language caption". ACL. 2017.

Subject-Verb-Object Probes

- SVO-Probes: subject-verb-object sentences with a focus on verbs
- Binary classification setup similar to FOIL
- Models largely fail to distinguish images with fine-grained verb differences
- Accuracy below chance on negative pairs
- Verb understanding is harder than subject or object understanding



Hendricks, Lisa Anne, and Aida Nematzadeh. "Probing Image-Language Transformers for Verb Understanding." Findings of ACL-IJCNLP 2021.

Winoground

- 1,600 text-image pairs to evaluate compositional understanding
- Examples curated by human experts to assess visual grounding skills
  - Curated in such a way that captions have the same words but appear in different order
- Task: match the correct caption to its image
- Models struggle, often performing below chance

(a) there is [a mug] in [some grass]

(c) a person [sits] and a dog [stands]

(b) there is [some grass] in [a mug]

(d) a person [stands] and a dog [sits]

Thrush et al. (2022). Winoground: Probing vision and language models for visio-linguistic compositionality. CVPR.

Multi-task Evaluation

- New category of benchmark that focuses on assessing the abilities of general-purpose models
  - General-purpose models should be evaluated across a wide range of capabilities!
- How do we do this? By generating or aggregating tasks from different families with the aim to maximise coverage over different capabilities
- These datasets are typically very large and require extensive computational resource; used by frontier labs to showcase their models!
- However, always pay attention to the trade-off between scale and quality of the datasets

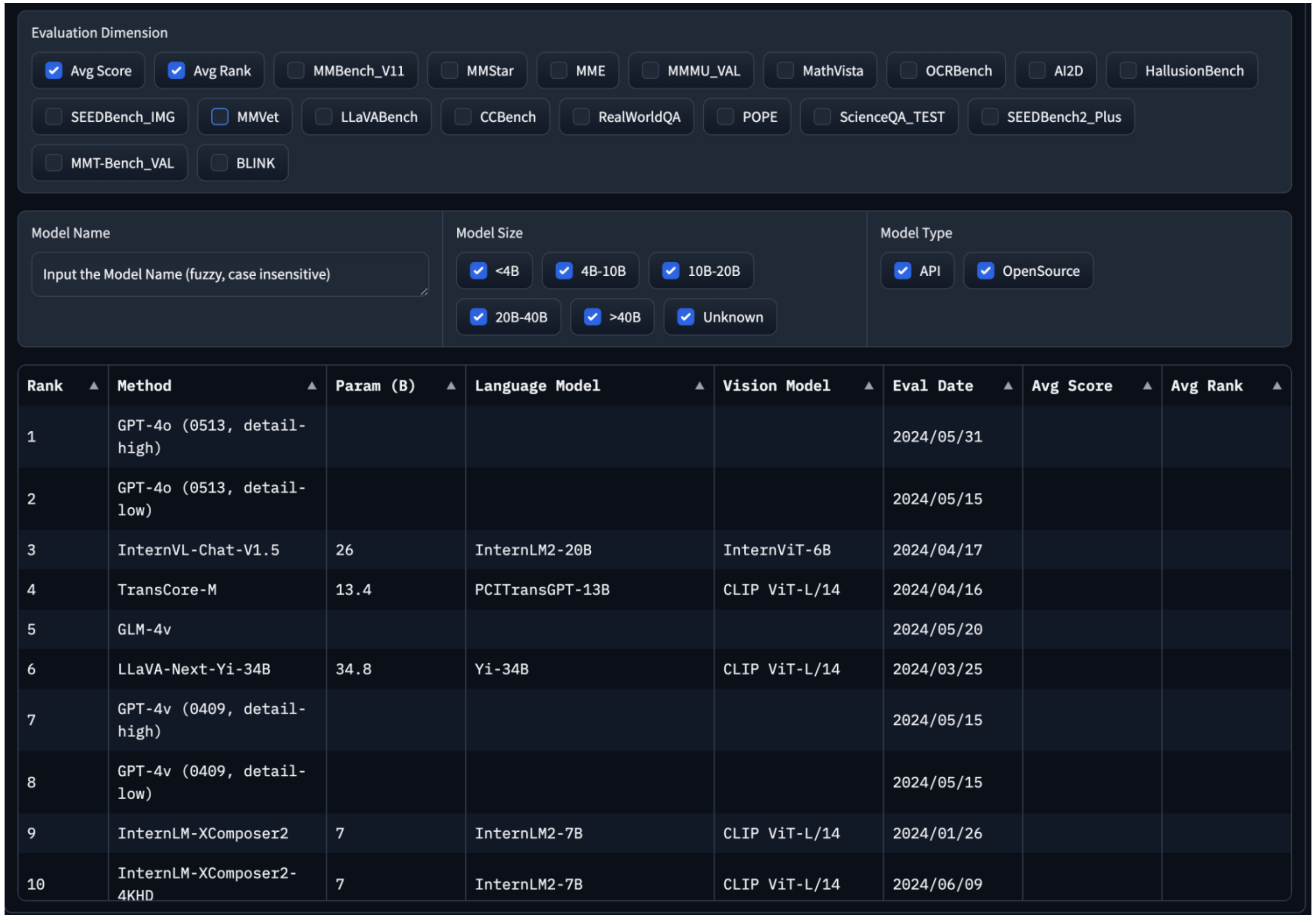
VLMEvalKit: A Toolkit for Evaluating Large Vision-Language Models

VLMEvalKit

- Provides a user-friendly and comprehensive framework for researchers and developers to evaluate existing VLMs and publish reproducible evaluation results
- Over 200 different VLMs including both proprietary APIs and open-weight models
- Evaluates a broad range of capabilities such as:
  - Document Understanding
  - Diagram and chart understanding
  - Image captioning
  - and many more included in more than 80 different multi-modal benchmarks
- Available on Github: https://github.com/open-compass/VLMEvalKit

VLMEvalKit

A Toolkit for Evaluating Large Vision-Language Models

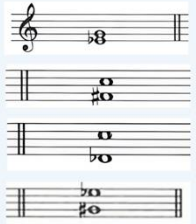
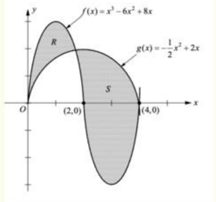
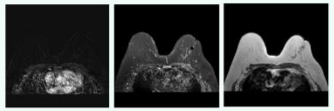
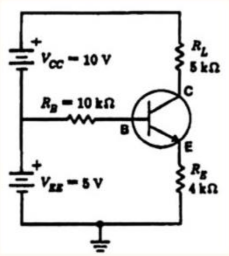


Rank	Method	Param (B)	Language Model	Vision Model	Eval Date	Avg Score	Avg Rank
1	GPT-4o (0513, detail-high)				2024/05/31		
2	GPT-4o (0513, detail-low)				2024/05/15		
3	InternVL-Chat-V1.5	26	InternLM2-20B	InternViT-6B	2024/04/17		
4	TransCore-M	13.4	PCITransGPT-13B	CLIP ViT-L/14	2024/04/16		
5	GLM-4v				2024/05/20		
6	LLaVA-Next-Yi-34B	34.8	Yi-34B	CLIP ViT-L/14	2024/03/25		
7	GPT-4v (0409, detail-high)				2024/05/15		
8	GPT-4v (0409, detail-low)				2024/05/15		
9	InternLM-XComposer2	7	InternLM2-7B	CLIP ViT-L/14	2024/01/26		
10	InternLM-XComposer2-4KHD	7	InternLM2-7B	CLIP ViT-L/14	2024/06/09		

https://huggingface.co/spaces/opencompass/open_vlm_leaderboard

MMMU

A Massive Multi-discipline Multimodal Understanding and Reasoning Benchmark for Expert AGI

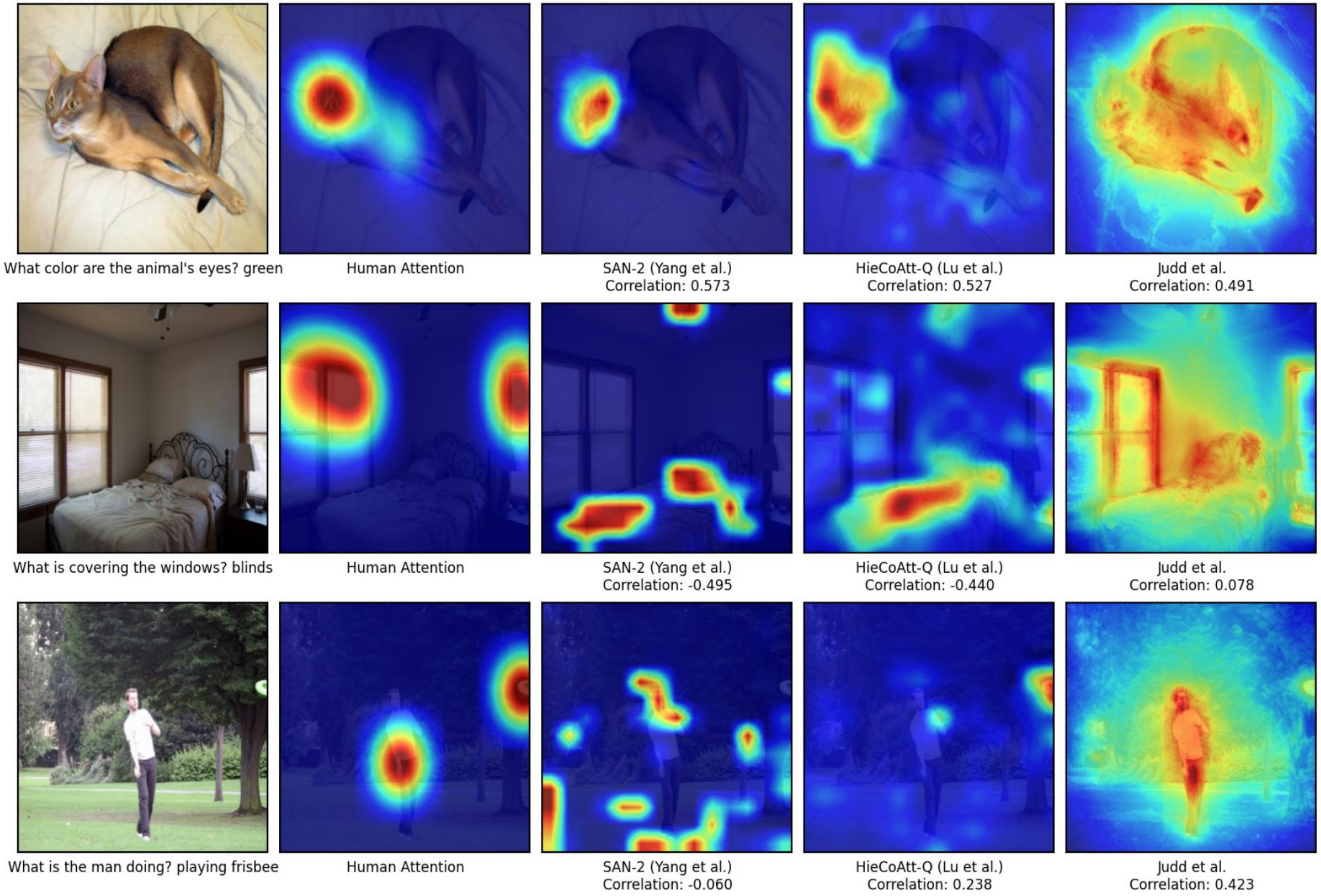
Art & Design	Business	Science
Question: Among the following harmonic intervals, which one is constructed incorrectly? **Options:** (A) Major third <*image 1*> (B) Diminished fifth <*image 2*> **(C) Minor seventh <*image 3*>** (D) Diminished sixth <*image 4*> 	**Question:** ...The graph shown is compiled from data collected by Gallup <*image 1*>. Find the probability that the selected Emotional Health Index Score is between 80.5 and 82? **Options:** (A) 0 (B) 0.2142 **(C) 0.3571** (D) 0.5 	**Question:** <*image 1*> The region bounded by the graph as shown above. Choose an integral expression that can be used to find the area of R. **Options:** **(A)** $\int_0^{1.5}[f(x)-g(x)]dx$ (B) $\int_0^{1.5}[g(x)-f(x)]dx$ (C) $\int_0^{2}[f(x)-g(x)]dx$ (D) $\int_0^{2}[g(x)-x(x)]dx$ 
Subject: Music; **Subfield:** Music; **Image Type:** Sheet Music; **Difficulty:** Medium	**Subject:** Marketing; **Subfield:** Market Research; **Image Type:** Plots and Charts; **Difficulty:** Medium	**Subject:** Math; **Subfield:** Calculus; **Image Type:** Mathematical Notations; **Difficulty:** Easy
Health & Medicine	**Humanities & Social Science**	**Tech & Engineering**
Question: You are shown subtraction <*image 1*>, T2 weighted <*image 2*> and T1 weighted axial <*image 3*> from a screening breast MRI. What is the etiology of the finding in the left breast? **Options:** (A) Susceptibility artifact (B) Hematoma **(C) Fat necrosis** (D) Silicone granuloma 	**Question:** In the political cartoon, the United States is seen as fulfilling which of the following roles? <*image 1*> **Option:** (A) Oppressor (B) Imperialist **(C) Savior** (D) Isolationist 	**Question:** Find the VCE for the circuit shown in <*image 1*>. Neglect VBE **Answer:** 3.75 **Explanation:** ...IE = [(VEE) / (RE)] = [(5 V) / (4 k-ohm)] = 1.25 mA; VCE = VCC - IERL = 10 V - (1.25 mA) 5 k-ohm; VCE = 10 V - 6.25 V = 3.75 V 
Subject: Clinical Medicine; **Subfield:** Clinical Radiology; **Image Type:** Body Scans: MRI, CT.; **Difficulty:** Hard	**Subject:** History; **Subfield:** Modern History; **Image Type:** Comics and Cartoons; **Difficulty:** Easy	**Subject:** Electronics; **Subfield:** Analog electronics; **Image Type:** Diagrams; **Difficulty:** Hard

Yue, Xiang, et al. "Mmmu: A massive multi-discipline multimodal understanding and reasoning benchmark for expert agi." CVPR. 2024.

Intrinsic Evaluation

- By grounding language into vision, arguably multimodal models have a representational advantage over text-only models
- We therefore need ways to evaluate the quality of the representations learned by the models
- For example:
  – Do they learn representations that better align with human multimodal knowledge and processing?
  – Can we improve them by studying their inner workings?
- Mechanistic interpretability for VLMs is still underexplored and there is a lot to be done to understand how both modalities are represented in modern models

Comparing attention patterns in VQA



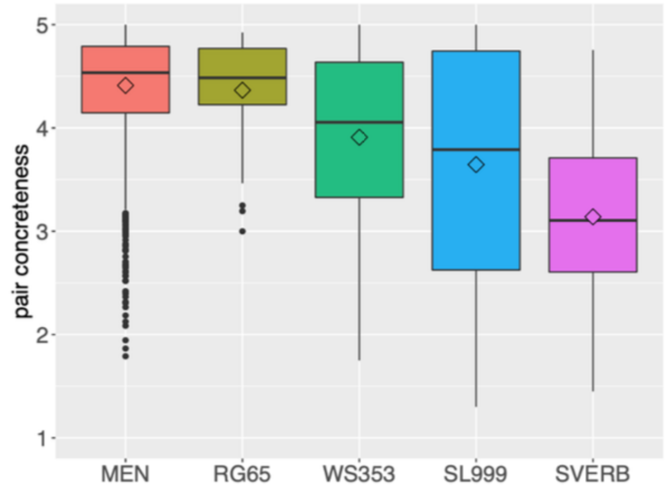
Das, Abhishek, et al. "Human attention in visual question answering: Do humans and deep networks look at the same regions?." Computer Vision and Image Understanding (2017)

Representational quality: correlation with semantic similarity judgements

- Do multimodal pre-trained models learn representations that are more aligned with human semantic similarity judgements?

The level of concreteness of the words being judged varies per dataset

Very concrete:
doughnut, muffin



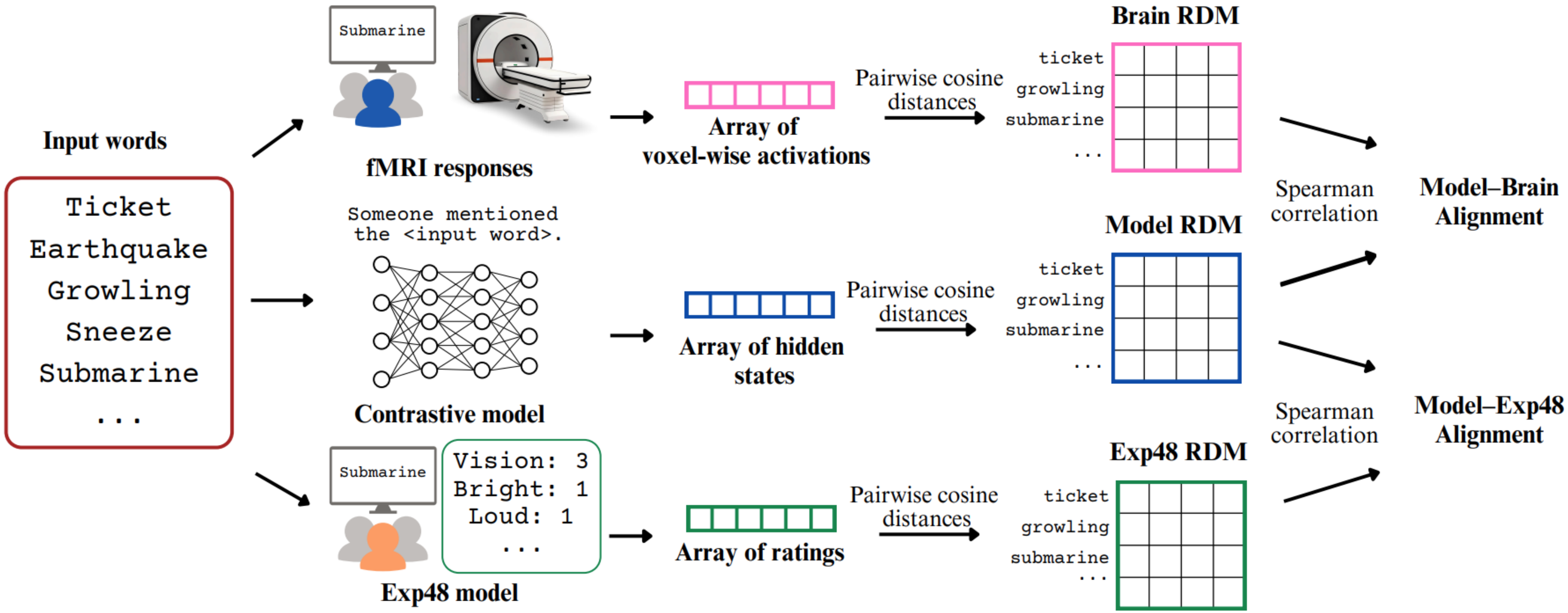
Very abstract:
freedom, dreams

Multimodal models are better than text-only ones at approximating similarity judgements of concrete words

Sandro Pezzelle, Ece Takmaz, and Raquel Fernández. "Word representation learning in multimodal pre-trained transformers: An intrinsic evaluation." TACL (2021).

Representational quality: correlation with brain responses

- Do multimodal pre-trained models learn representations that are more aligned with how the brain represents conceptual knowledge?



Anna Bavaresco and Raquel Fernández. "Experiential Semantic Information and Brain Alignment: Are Multimodal Models Better than Language Models?." CoNLL (2025).

Summary

- Current NLP models have been trained using large-scale datasets only. This limits the variety of signal they can access to for learning.
- It has been shown that language understanding is an activity that leverages multiple parts of the brains that are involved to complete other tasks as well. The brain is inherently multimodal!
- How do we design Perceptually grounded NLP systems? NLP systems can live in different world scopes — different data distributions that an agent can leverage to learn language
- Vision+Language models are a promising avenue to bridge the gap between fully multimodal systems and text-only models

Homework

- Use this slide deck to revise the main concepts that we covered today and Suglia, Alessandro, Ioannis Konstas, and Oliver Lemon. "Visually grounded language learning: a review of language games, datasets, tasks, and models." *Journal of Artificial Intelligence Research* 79 (2024): 173-239.
  - Section 1, 2, 3
- Focus your attention on:
  - Why do we care about multimodal models beyond applications?
  - The overall training regime to create VLMs
  - Once we have a VLM, how do we evaluate it? And for what task?
- Next lecture will focus on how frontier labs design and train state-of-the-art Vision and Language Models